



## High Level Problems (HLP)

- If the equation  $z^4 + a_1z^3 + a_2z^2 + a_3z + a_4 = 0$  where  $a_1, a_2, a_3, a_4$  are real coefficient different from zero, has a purely imaginary root, then find the value of  $\frac{a_3}{a_1a_2} + \frac{a_1a_4}{a_2a_3}$
- If  $|z_1| = 2, |z_2| = 3, |z_3| = 4$  and  $|2z_1 + 3z_2 + 4z_3| = 4$ , then find the value of  $|8z_2z_3 + 27z_3z_1 + 64z_1z_2|$
- If  $|z|^2 + \bar{A}z^2 + A\bar{z}^2 + B\bar{z} + \bar{B}z + c = 0$  represents a pair of intersecting lines with angle of intersection ' $\theta$ ' then find the value of  $|A|$
- If  $z^2 + \alpha z + \beta = 0$  ( $\alpha, \beta$  are complex numbers) has a real root then prove that  $(\bar{\alpha} - \alpha)(\alpha\bar{\beta} - \bar{\alpha}\beta) = (\beta - \bar{\beta})^2$
- If  $z_1, z_2, z_3$  be three complex number such that  $|z_1| = |z_2| = |z_3| = 1$  and  $\frac{z_1^2}{z_2z_3} + \frac{z_2^2}{z_1z_3} + \frac{z_3^2}{z_1z_2} + 1 = 0$  then sum of all the possible values of  $|z_1 + z_2 + z_3|$
- Number of complex number ( $z$ ) satisfying  $|z|^2 = |z|^{n-2}z^2 + |z|^{n-2}\bar{z} + 1$  such that  $\text{Re}(z) \neq -\frac{1}{2}$  and  $n = 2\lambda + 1, \lambda \in \mathbb{N}$ .
- Let  $z_1$  &  $z_2$  be any two arbitrary complex numbers then prove that  
 (i)  $|z_1 + z_2| = \left| \frac{z_1}{|z_1|} |z_2| + \frac{z_2}{|z_2|} |z_1| \right|$  (ii)  $|z_1 + z_2| \geq \frac{1}{2} (|z_1| + |z_2|) \left| \frac{z_1}{|z_1|} + \frac{z_2}{|z_2|} \right|$
- Prove that  
 (i)  $\left| \frac{z}{|z|} - 1 \right| \leq |\arg z|$ . (ii)  $|z - 1| \leq ||z| - 1| + |z| |\arg z|$ .
- Prove that  $|\text{Im}(z^n)| \leq n |\text{Im}(z)| |z|^{n-1}, n \in \mathbb{I}^+$
- If  $|z-1| + |z+3| \leq 8$  then find the range of values of  $|z-4|$ .
- Show that all the roots of the equation  $a_1z^3 + a_2z^2 + a_3z + a_4 = 3$ , where  $|a_i| \leq 1, i = 1, 2, 3, 4$  lie outside the circle with centre origin and radius  $2/3$ .
- Consider the locus of the complex number  $z$  in the Argand plane is given by  $\text{Re}(z) - 2 = |z - 7 + 2i|$ . Let  $P(z_1)$  and  $Q(z_2)$  be two complex number satisfying the given locus and also satisfying  $\arg \left( \frac{z_1 - (2 + \alpha i)}{z_2 - (2 + \alpha i)} \right) = \frac{\pi}{2} (\alpha \in \mathbb{R})$  then find the minimum value of  $PQ$
- Find the mirror image of the curve  $\left| \frac{z - z_1}{z - z_2} \right| = a, a \in \mathbb{R}^+ a \neq 1$  about the line  $|z - z_1| = |z - z_2|$ .





14. Let  $z_1$  and  $z_2$  are the two complex numbers satisfying  $|z - 3 - 4i| = 3$ . Such that  $\text{Arg}\left(\frac{z_1}{z_2}\right)$  is maximum then find the value of  $|z_1 - z_2|$ .
15. If  $z_1$  and  $z_2$  are the two complex numbers satisfying  $|z - 3 - 4i| = 8$  and  $\text{Arg}\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$  then find the range of the values of  $|z_1 - z_2|$ .
16. If  $|z - z_1| = |z_1|$  and  $|z - z_2| = |z_2|$  be the two circles and the two circles touch each other then prove that  $\text{Im}\left(\frac{z_1}{z_2}\right) = 0$ .
17. If  $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$ ; where  $p, q, r$  are the modulus of non-zero complex numbers  $u, v, w$  respectively, prove that,  $\arg \frac{w}{v} = \arg \left( \frac{w-u}{v-u} \right)^2$ .
18. If  $|z_2 + iz_1| = |z_1| + |z_2|$  and  $|z_1| = 3$  &  $|z_2| = 4$ , if affix of A, B, C are  $z_1, z_2, \left(\frac{z_2 - iz_1}{1-i}\right)$  respectively. Then find the area of  $\triangle ABC$ .
19. Find the locus of mid-point of line segment intercepted between real and imaginary axes, by the line  $a\bar{z} + \bar{a}z + b = 0$ , where 'b' is real parameter and 'a' is a fixed complex number such that  $\text{Re}(a) \neq 0$ ,  $\text{Im}(a) \neq 0$ .
20. Given  $z_1 + z_2 + z_3 = A$ ,  $z_1 + z_2\omega + z_3\omega^2 = B$ ,  $z_1 + z_2\omega^2 + z_3\omega = C$ , where  $\omega$  is cube root of unity,  
 (a) express  $z_1, z_2, z_3$  in terms of A, B, C.  
 (b) prove that,  $|A|^2 + |B|^2 + |C|^2 = 3(|z_1|^2 + |z_2|^2 + |z_3|^2)$ .  
 (c) prove that  $A^3 + B^3 + C^3 - 3ABC = 27z_1z_2z_3$
21. If  $w \neq 1$  is  $n^{\text{th}}$  root of unity, then find the value of  $\sum_{k=0}^{n-1} |z_1 + w^k z_2|^2$ .
22. Let a, b, c be distinct complex numbers such that  $\frac{a}{1-b} = \frac{b}{1-c} = \frac{c}{1-a} = k$ , ( $a, b, c \neq 1$ ). Find the value of k.
23. If  $\alpha = e^{\frac{2\pi i}{7}}$  and  $f(x) = A_0 + \sum_{k=1}^{20} A_k x^k$ , then find the value of,  $f(x) + f(\alpha x) + \dots + f(\alpha^{20}x)$  independent of  $\alpha$ .
24. Given,  $z = \cos \frac{2\pi}{2n+1} + i \sin \frac{2\pi}{2n+1}$ , 'n' a positive integer, find the equation whose roots are,  $\alpha = z + z^3 + \dots + z^{2n-1}$  and  $\beta = z^2 + z^4 + \dots + z^{2n}$ .
25. Prove that  $\cos\left(\frac{2\pi}{2n+1}\right) + \cos\left(\frac{4\pi}{2n+1}\right) + \cos\left(\frac{6\pi}{2n+1}\right) + \dots + \cos\left(\frac{2n\pi}{2n+1}\right) = -\frac{1}{2}$  When  $n \in \mathbb{N}$ .





26. Proof that (i)  $\sin \frac{\pi}{2k+1} \sin \frac{2\pi}{2k+1} \dots \sin \frac{k\pi}{2k+1} = \frac{\sqrt{2k+1}}{2^k}$   
 (ii)  $\cos \frac{\pi}{2k+1} \cos \frac{2\pi}{2k+1} \dots \cos \frac{k\pi}{2k+1} = \frac{1}{2^k}$
27. If  $Z_r, r = 1, 2, 3, \dots, 2m, m \in \mathbb{N}$  are the roots of the equation  $Z^{2m} + Z^{2m-1} + Z^{2m-2} + \dots + Z + 1 = 0$ , then prove that  $\sum_{r=1}^{2m} \frac{1}{Z_r - 1} = -m$
28. The points represented by the complex numbers  $a, b, c$  lie on a circle with centre  $O$  and radius  $r$ . The tangent at  $c$  cuts the chord joining the points  $a, b$  at  $z$ . Show that  $z = \frac{a^{-1} + b^{-1} - 2c^{-1}}{a^{-1}b^{-1} - c^{-2}}$
29. Show that for the given complex numbers  $z_1$  and  $z_2$  and for a real constant  $c$  the equation  $(z_1 + \lambda z_2)\bar{z} + (\bar{z}_1 + \lambda \bar{z}_2)z + c = 0$  represents a family of concurrent lines and also find the fixed point of the family. (where  $\lambda$  is a real parameter)
30. Let  $z_1, z_2, z_3$  are three pair wise distinct complex numbers and  $t_1, t_2, t_3$  are non-negative real numbers such that  $t_1 + t_2 + t_3 = 1$ . Prove that the complex number  $z = t_1 z_1 + t_2 z_2 + t_3 z_3$  lies inside a triangle with vertices  $z_1, z_2, z_3$  or on its boundary.

## Answers

1. 1    2. 96    3.  $\frac{\sec \theta}{2}$     5. 3    6. 2    10.  $[1, 9]$
12. 10    13.  $\left| \frac{z - z_2}{z - z_1} \right| = a$     14.  $\frac{24}{5}$     15.  $[\sqrt{103} - 5, \sqrt{103} + 5]$
18.  $\frac{25}{4}$     19.  $\bar{a}z + az = 0$     20. (a)  $z_1 = \frac{A + B + C}{3}, z_2 = \frac{A + B\omega^2 + C\omega}{3}, z_3 = \frac{A + B\omega + C\omega^2}{3}$
21.  $n(|z_1|^2 + |z_2|^2)$     22.  $-\omega$  or  $-\omega^2$     23.  $7A_0 + 7A_7x^7 + 7A_{14}x^{14}$
24.  $z^2 + z + \frac{\sin^2 n\theta}{\sin^2 \theta} = 0$ , where  $\theta = \frac{2\pi}{2n+1}$     29.  $z = \frac{cz_2}{z_1\bar{z}_2 - z_2\bar{z}_1}$